

基於時間與價格波動的自動動態動量保險庫

Automated Dynamic Momentum Vault

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Goal of smart contract

- 自動避險機制：
 - 當市場處於行情好 ($\text{market} \geq 800$ BULL) 時，資金將100%存入虛擬貨幣中。
 - 當市場處於保守 ($501 < \text{market} < 799$ BEAR) 時，資金將80%存入虛擬貨幣中，降低用戶資產在熊市的風險暴露。
 - 當市場處於閃崩斷路器 ($\text{market} \leq 500$ FLASH_CRASH)，存入 / 提款功能將會被鎖定無法使用。另外開啟緊急保本退款，退款數目將全數退回，而不扣手續費及不參與分紅池活動。

Goal of smart contract cont.

- 時間鎖、分紅池機制：
 - 當需提款存入不到 3 天的款項時，須被強製扣除 5% 的罰款並存入分紅池中。
 - 當提款已存入超過 3 天的款項，可按比例瓜分分紅池內獎金。
 - 合約計算分紅公式：

$$bonus = \frac{aCCUmIatedPenaltyPool \times user.shares}{totalVaultShares}$$

The function of smart contract

- `owner` 合約管理員：擁有更改時間及市場價格的權限。
- `currentMarketState` state machine: 包含BULL（牛市）、BEAR（熊市）、FLASH_CRASH（閃崩）。合約所有的業務邏輯（存、取款）都會根據這個狀態自動切換行為。
- `CRASH\_THRESHOLD` = 500：閃崩臨界值。只要預言機回報的價格低於或等於此數值，將直接觸發斷路器。
- `EARLY\_WITHDRAW\_PENALTY\_TIME` = 3 days：時間鎖門檻。存入資產未滿3天者將面臨早退懲罰。
- `accumulatedPenaltyPool`（分紅池）：用來動態記錄和儲存所有「未滿3天提款者」被扣除的5% 罰款總額。
- `UserInfo`（用戶帳本）：精準記錄每位用戶當前的份額餘額（shares）以及最後一次存款的虛擬時間戳記（depositTimestamp）。

The function of smart contract cont.

- ``setMarketPrice(uint256 _newPrice)`` 模擬設定市場價格，及狀態BULL / BEAR / FLASH_CRASH，僅限admin。
- ``deposit()`` 存款：非崩盤下 (whenNotCrashed) 才能使用。
 - 牛市（BULL）：按 100% 比例足額發行。存入 1 ETH 獲得 1 份額。
 - 熊市（BEAR）：按 80% 比例保守發行。存入 1 ETH 僅獲得 0.8 份額。剩餘的 20% 資金安全地留存在合約總資產（`address(this).balance`）中作為流動性緩衝金。
- ``withdraw(uint256 _sharesToWithdraw)`` 提款：非崩盤下 (whenNotCrashed) 才能使用。
 - 未滿 3 天（閃電提領）：合約判定用戶缺乏耐心，會針對本次提領的數量強制扣除 5% 罰款（`penalty`），並直接記錄累加進 `accumulatedPenaltyPool`（分紅池）。
 - 滿 3 天以上（長期持有）：享有 0 手續費福利，且能按本次提領份額佔全場的比例，精準瓜分當前分紅池裡的獎勵（`bonus`）。
 - 記帳變動：`totalVaultShares`（全站總份額）與用戶的 `user.shares` 僅扣除本次提領的部分，其餘資產繼續留在合約中增值。

The function of smart contract cont.

- `emergencyRefund()` 緊急保本退款通道：狀態處於 ``FLASH_CRASH`` 才會出現。讓所有人可以無視 3 天時間鎖，無條件、零手續費地將當前剩餘的所有份額（100% 原始本金）安全撤離。
- ``cheatForwardTime(uint256 _seconds)`` 測試用時間調整器：僅限管理員。當調用此函數時（例如快進 4 天），合約內部所有的時間戳計算（`getVirtualTimestamp()`）都會自動往後飛躍 4 天。

Snapshot of all step: Set market state

- BULL

DYNAMIC MOMENTUM VAULT

0xf39f...2266

虛擬預言機市價
\$1200

保險庫狀態
BULL (牛市狂歡)

您的份額 (SHARES)
0.0

累積罰款分紅池
0.0 ETH

開發者控制台
OWNER ONLY

動態調整市場報價
\$1200

向區塊鏈廣播新價格

區塊鏈時光旅行調節器
讓區塊鏈時間瞬間快進 4 天！

點擊此按鈕將發送低階 RPC 指令至 Hardhat 本地節點，模擬存錢超過 3 天後的場景。

基金核心互動介面

存入 ETH 參與智能調配
0.0 ETH
立即存入資產

贖回資產 (自動結算分紅/懲罰)
提示：存入不滿 3 天提款將扣除 5% 注入分紅池；滿 3 天可瓜分分紅池。
輸入提款份額 (Shares)
申請安全提款

MetaMask

Imported Account 1
0xf39Fd...92266

US\$17,723,697.40
+US\$0.00 (+0.00%) Discover

Buy Swap 發送 接收

7 days left for \$50K
There's still time to join the competition and climb the...

Tokens Perps DeFi NFTs 交易紀錄

Hardhat Local
Jun 16, 2026

Sent
已確認
-0 ETH
-US\$0.00

Withdraw
已確認
-0 ETH
-US\$0.00

Deposit
已確認
-100 ETH
-US\$17,237.00

localhost:5500 Imported Account 1

Snapshot of all step: Set market state cont.

- BEAR

DYNAMIC MOMENTUM VAULT

0xf39f...2266

虛擬預言機市價
\$750

保險庫狀態
BEAR (熊市保守)

您的份額 (SHARES)
0.0

累積罰款分紅池
0.0 ETH

開發者控制台 **OWNER ONLY**

動態調整市場報價 **\$750**

向區塊鏈廣播新價格

區塊鏈時光旅行調節器

讓區塊鏈時間瞬間快進 4 天!

點擊此按鈕將發送低階 RPC 指令至 Hardhat 本地節點，模擬存錢超過 3 天後的場景。

基金核心互動介面

存入 ETH 參與智能調配

0.0 ETH

立即存入資產

贖回資產 (自動結算分紅/懲罰)

提示：存入不滿 3 天提款將扣除 5% 注入分紅池；滿 3 天可瓜分分紅池。

輸入提款份額 (Shares)

申請安全提款

MetaMask

Imported Account 1

0xf39Fd...92266

US\$17,723,697.31
+US\$0.00 (+0.00%) [Discover](#)

Buy

Swap

發送

接收

7 days left for \$50K

There's still time to join the competition and climb the...

Tokens

Perps

DeFi

NFTs

交易紀錄

Hardhat Local

Jun 16, 2026

Sent

已確認

-0 ETH

-US\$0.00

Withdraw

已確認

-0 ETH

-US\$0.00

Deposit

已確認

-100 ETH

-US\$177,237.00

localhost:5500

Imported Account 1

Snapshot of all step: Set market state cont.

- FLASH_CRASH & 緊急保本退款

DYNAMIC MOMENTUM VAULT

0xf39f...2266

虛擬預言機市價
\$400

保險庫狀態
FLASH_CRASH
(閃崩斷路器!)

您的份額 (SHARES)
0.0

累積罰款分紅池
0.0 ETH

開發者控制台
OWNER ONLY

動態調整市場報價
\$400

向區塊鏈廣播新價格

區塊鏈時光旅行調節器
讓區塊鏈時間瞬間快進 4 天!

點擊此按鈕將發送低隨 RPC 指令至 Hardhat 本地節點，模擬存錢超過 3 天後的場景。

基金核心互動介面

存入 ETH 參與智能調配
0.0 ETH
立即存入資產

贖回資產 (自動結算分紅/懲罰)
提示: 存入不滿 3 天提款將扣除 5% 注入分紅池; 滿 3 天可瓜分分紅池。
輸入提款份額 (Shares)
申請安全提款
觸發緊急保本退款

MetaMask

Imported Account 1
0xf39Fd...92266

US\$17,723,697.21
+US\$0.00 (+0.00%) Discover

Buy Swap 發送 接收

7 days left for \$50K
There's still time to join the competition and climb the...

Tokens Perps DeFi NFTs 交易紀錄

Hardhat Local
Jun 16, 2026

Sent
已確認

-0 ETH
-US\$0.00

Withdraw
已確認

-0 ETH
-US\$0.00

Deposit
已確認

-100 ETH
-US\$177,237.00

localhost:5500 Imported Account 1

Snapshot of all step: Deposit 100\$

- Market state: BULL, save 100% in account.

The image displays two side-by-side screenshots from a web browser. The left screenshot shows the 'DYNAMIC MOMENTUM VAULT' interface. It features a top navigation bar with the vault's name and a wallet address '0xf39f...2266'. Below this, there are four main status boxes: '虛擬預言機市價' (Virtual Oracle Market Price) at \$1000, '保險庫狀態' (Vault Status) as 'BULL (牛市狂歡)' (Bull Market Madness), '您的份額 (SHARES)' (Your Shares) at 100.0 (highlighted with a red box), and '累積罰款分紅池' (Accumulated Penalty Dividend Pool) at 0.0 ETH. A '開發者控制台' (Developer Console) section on the right includes an 'OWNER ONLY' toggle, a market price slider set to \$1000, and a '向區塊鏈廣播新價格' (Broadcast New Price to Blockchain) button. The bottom section, '基金核心互動介面' (Fund Core Interaction Interface), contains two main areas: '存入 ETH 參與智能調配' (Deposit ETH to Participate in Smart Allocation) with a '0.0 ETH' input and an '立即存入資產' (Deposit Assets Immediately) button, and '贖回資產 (自動結算分紅/懲罰)' (Redeem Assets (Automatic Settlement of Dividend/Penalty)) with a '申請安全提款' (Apply for Safe Withdrawal) button. The right screenshot shows the MetaMask mobile app interface. It displays 'Imported Account 1' with a balance of 'US\$17,723,697.13'. Below the balance are buttons for 'Buy', 'Swap', '發送' (Send), and '接收' (Receive). A notification banner at the top right says '7 days left for \$50K'. The bottom section, '交易紀錄' (Transaction History), lists recent transactions: 'Sent' (-0 ETH), 'Withdraw' (-0 ETH), and 'Deposit' (-100 ETH, confirmed, highlighted with a red box). The bottom status bar shows 'localhost:5500' and 'Imported Account 1'.

DYNAMIC MOMENTUM VAULT (0xf39f...2266)

- 虛擬預言機市價: **\$1000**
- 保險庫狀態: **BULL (牛市狂歡)**
- 您的份額 (SHARES): **100.0**
- 累積罰款分紅池: **0.0 ETH**

開發者控制台 (OWNER ONLY)

- 動態調整市場報價: **\$1000**
- 向區塊鏈廣播新價格

基金核心互動介面

- 存入 ETH 參與智能調配**: 0.0 ETH, **立即存入資產**
- 贖回資產 (自動結算分紅/懲罰)**: 提示: 存入不滿 3 天提款將扣除 5% 注入分紅池; 滿 3 天可瓜分分紅池。 **申請安全提款**

MetaMask (Imported Account 1, 0xf39Fd...92266)

US\$17,723,697.13 (+US\$0.00 (+0.00%) [Discover](#))

Buy Swap 發送 接收

7 days left for \$50K. There's still time to join the competition and climb the...

Tokens Perps DeFi NFTs **交易紀錄**

Hardhat Local

Jun 16, 2026

- Sent 已確認 -0 ETH -US\$0.00
- Withdraw 已確認 -0 ETH -US\$0.00
- Deposit 已確認 -100 ETH -US\$17,237.00**

localhost:5500 Imported Account 1

Snapshot of all step: Deposit 100\$ cont.

- Market state: BEAR, save 80% in account.

The image displays two side-by-side screenshots. The left screenshot shows the 'DYNAMIC MOMENTUM VAULT' interface with a dark theme. It features several key metrics: '虛擬預言機市價' at \$650, '保險庫狀態' as BEAR (熊市保守), '您的份額 (SHARES)' at 80.0 (highlighted with a red box), and '累積罰款分紅池' at 5.0 ETH. Below these are sections for '基金核心互動介面' with options to '存入 ETH' (0.0 ETH) or '贖回資產', and a '開發者控制台' with an 'OWNER ONLY' toggle. The right screenshot is a MetaMask mobile app interface showing 'Imported Account 1' with a balance of US\$17,537,597.60. It includes buttons for 'Buy', 'Swap', '發送', and '接收'. A notification states '7 days left for \$50K'. The '交易紀錄' (Transaction History) section is active, showing a list of transactions. The 'Deposit' transaction is highlighted with a red box, showing '-100 ETH' and '-US\$177,237.00' with a status of '已確認'.

DYNAMIC MOMENTUM VAULT (0xf39f...2266)

- 虛擬預言機市價: \$650
- 保險庫狀態: BEAR (熊市保守)
- 您的份額 (SHARES): 80.0
- 累積罰款分紅池: 5.0 ETH

基金核心互動介面

- 存入 ETH 參與智能調配: 0.0 ETH, 立即存入資產
- 贖回資產 (自動結算分紅/懲罰): 提示: 存入不滿 3 天提款將扣除 5% 注入分紅池; 滿 3 天可瓜分分紅池。 輸入提款份額 (Shares), 申請安全提款

開發者控制台 (OWNER ONLY)

- 動態調整市場報價: \$650, 向區塊鏈廣播新價格
- 區塊鏈時光旅行調節器: 讓區塊鏈時間瞬間快進 4 天!

MetaMask (Imported Account 1, 0xf39Fd...92266)

US\$17,537,597.60 (+US\$0.00 (+0.00%) Discover)

Buy Swap 發送 接收

7 days left for \$50K: There's still time to join the competition and climb the...

Tokens Perps DeFi NFTs 交易紀錄

Hardhat Local Jun 16, 2026

Transaction	Amount	Status
Sent	-0 ETH	已確認
Withdraw	-0 ETH	已確認
Deposit	-100 ETH	已確認

localhost:5500 Imported Account 1

Snapshot of all step: Withdraw within 3 days

- Withdraw within 3 days with penalty: before withdraw

DYNAMIC MOMENTUM VAULT0xf39f...2266

虛擬預言機市價
\$1000

保險庫狀態
BULL (牛市狂歡)

您的份額 (SHARES)
100.0

🔥 累積罰款分紅池
9.0 ETH

⚡ 基金核心互動介面

存入 ETH 參與智能調配
0.0 ETH
立即存入資產

贖回資產 (自動結算分紅/懲罰)
提示: 存入不滿 3 天提款將扣除 5% 注入分紅池;
滿 3 天可瓜分分紅池。
50
申請安全提款

🔧 開發者控制台OWNER ONLY

動態調整市場報價\$1000
向區塊鏈廣播新價格

🕒 區塊鏈時光旅行調節器
⚡ 讓區塊鏈時間瞬間快進 4 天!
點擊此按鈕將發送低階 RPC 指令至 Hardhat 本地節點, 模擬存錢超過 3 天後的場景。

Snapshot of all step: Withdraw within 3 days cont.

- Withdraw within 3 days with penalty: after withdraw

DYNAMIC MOMENTUM VAULT0xf39f...2266

虛擬預言機市價
\$1000

保險庫狀態
BULL (牛市狂歡)

您的份額 (SHARES)
50.0

累積罰款分紅池
11.5 ETH

開發者控制台OWNER ONLY

動態調整市場報價
\$1000
向區塊鏈廣播新價格

區塊鏈時光旅行調節器
讓區塊鏈時間瞬間快進 4 天!
點擊此按鈕將發送低階 RPC 指令至 Hardhat 本地節點，模擬存錢超過 3 天後的場景。

基金核心互動介面

存入 ETH 參與智能調配
0.0 ETH
立即存入資產

贖回資產 (自動結算分紅/懲罰)
提示: 存入不滿 3 天提款將扣除 5% 注入分紅池; 滿 3 天可瓜分分紅池。
輸入提款份額 (Shares)
申請安全提款

Snapshot of all step: Withdraw after 4 days cont.

- Withdraw after 4 days with **no** penalty: before withdraw

The screenshot displays the 'DYNAMIC MOMENTUM VAULT' interface. At the top, the title 'DYNAMIC MOMENTUM VAULT' is shown in green and blue, with a user address '0xf39f...2266' in a grey box on the right. Below the title, there are four main status cards: '虛擬預言機市價' (Virtual Oracle Market Price) at '\$1000', '保險庫狀態' (Vault Status) as 'BULL (牛市狂歡)' (Bull Market Madness), '您的份額 (SHARES)' (Your Share) at '100.0' (highlighted with a red box), and '累積罰款分紅池' (Accumulated Penalty Dividend Pool) at '11.5 ETH'. To the right is the '開發者控制台' (Developer Console) for 'OWNER ONLY', featuring a '動態調整市場報價' (Dynamic Market Price Adjustment) slider set to '\$1000' and a button '向區塊鏈廣播新價格' (Broadcast New Price to Blockchain). Below this is the '區塊鏈時光旅行調節器' (Blockchain Time Travel Regulator) section, which includes a button '讓區塊鏈時間瞬間快進 4 天!' (Let Blockchain Time Instantly Fast Forward 4 Days!) and a note about sending RPC commands to Hardhat. The bottom section, '基金核心互動介面' (Fund Core Interaction Interface), contains two main areas: '存入 ETH 參與智能調配' (Deposit ETH to Participate in Smart Allocation) with a '0.0 ETH' input and a green '立即存入資產' (Deposit Assets Immediately) button; and '贖回資產 (自動結算分紅/懲罰)' (Redeem Assets (Automatic Settlement of Dividend/Penalty)) with a '50' input and a button '申請安全提款' (Apply for Safe Withdrawal) (highlighted with a red box). A hint for redemption states: '提示：存入不滿 3 天提款將扣除 5% 注入分紅池；滿 3 天可瓜分分紅池。' (Note: If withdrawal is made within 3 days of deposit, 5% will be deducted and added to the dividend pool; after 3 days, you can share the dividend pool.)

Snapshot of all step: Withdraw after 4 days cont.

- Withdraw after 4 days with **no** penalty: after withdraw

The screenshot displays the 'DYNAMIC MOMENTUM VAULT' interface. At the top, the title 'DYNAMIC MOMENTUM VAULT' is on the left and the address '0xf39f...2266' is on the right. Below the title, there are four main status boxes: '虛擬預言機市價' (Virtual Oracle Market Price) at \$1000, '保險庫狀態' (Vault Status) as 'BULL (牛市狂歡)' (Bull Market Madness), '您的份額 (SHARES)' (Your Shares) at 50.0, and '累積罰款分紅池' (Accumulated Penalty Dividend Pool) at 5.75 ETH. The '50.0' and '5.75 ETH' values are highlighted with red boxes. To the right of these is the '開發者控制台' (Developer Console) for 'OWNER ONLY', featuring a '動態調整市場報價' (Dynamic Market Price Adjustment) slider set to \$1000, a '向區塊鏈廣播新價格' (Broadcast New Price to Blockchain) button, and a '區塊鏈時光旅行調節器' (Blockchain Time Travel Regulator) with a button to '讓區塊鏈時間瞬間快進 4 天!' (Fast forward blockchain time 4 days!). Below the main status boxes is the '基金核心互動介面' (Fund Core Interaction Interface) with two sections: '存入 ETH 參與智能調配' (Deposit ETH for intelligent allocation) with a '0.0 ETH' input and an '立即存入資產' (Deposit Assets Immediately) button, and '贖回資產 (自動結算分紅/懲罰)' (Redeem Assets (Automatic dividend/settlement/punishment)) with a prompt about 3-day rules, an input for '輸入提款份額 (Shares)' (Enter withdrawal amount in shares), and an '申請安全提款' (Apply for safe withdrawal) button.

● DYNAMIC MOMENTUM VAULT 0xf39f...2266

虛擬預言機市價
\$1000

保險庫狀態
BULL (牛市狂歡)

您的份額 (SHARES)
50.0

累積罰款分紅池
5.75 ETH

開發者控制台 OWNER ONLY

動態調整市場報價 **\$1000**

向區塊鏈廣播新價格

區塊鏈時光旅行調節器

讓區塊鏈時間瞬間快進 4 天!

點擊此按鈕將發送低階 RPC 指令至 Hardhat 本地節點，模擬存錢超過 3 天後的場景。

基金核心互動介面

存入 ETH 參與智能調配

0.0 ETH

立即存入資產

贖回資產 (自動結算分紅/懲罰)

提示：存入不滿 3 天提款將扣除 5% 注入分紅池；滿 3 天可瓜分分紅池。

輸入提款份額 (Shares)

申請安全提款

Contract problem: Liquid Trap

- 如果沒有人提早領錢出場，分紅池內的獎金將會0，也就是20%的錢將被鎖在合約裡，無人可領。
- 解決方法：
 - 引入淨值動態結算機制：（按比例退錢）
 - 你應得的錢 = $\frac{\text{你的份額}}{\text{總份額}} \times \text{合約目前總資產}$
 - 合約終局清算：
 - 當基金要關閉時，由 **admin** 呼叫，把合約裡殘留的、因為熊市少發份額而多出來的 **ETH**（現金儲備），全額提領出來，再手動發還給用戶，或當作基金經理人的操盤手續費（**Performance Fee**）。

Github Link

- <https://github.com/imlone1y/Automated-Dynamic-Momentum-Vault>